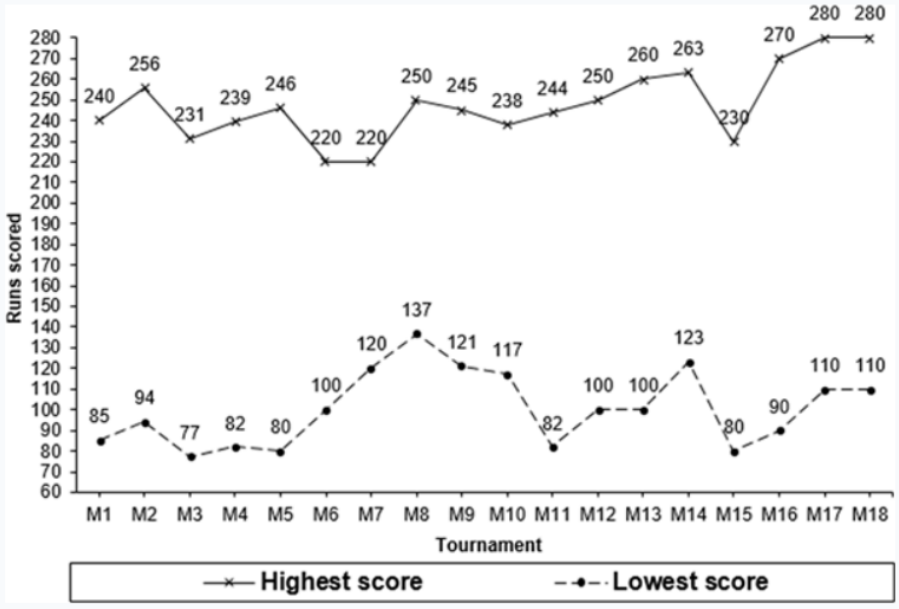
Directions for questions 1 to 5: Answer these questions on the basis of the information given below.

The line graph which gives the highest and the least runs scored by a regional cricket team in the last 18 tournaments played by it.



In which two consecutive tournaments is the difference between the highest and the lowest scores of the team, the same?

a) (M_{12}, M_{13})

b) (M_3, M_4)

c) (M_9, M_{10})

d) (M_{17}, M_{18}) (Your answer - ✓)

Solution:

In the tournaments M_{17} and M_{18}, the difference between the highest and the lowest scores is the same. Choice (D)

In which tournament is the ratio of the least score to the highest score of the team, the least?

a) M_3

b) M_5 (Your answer - ✓)

c) M_{16}

d) None of these

Solution:

If we compare the ratios of tournaments in M_3, M_5 and M_{16}

Tournaments	Ratio
M_3	$\frac{77}{231} = \frac{1}{3}$
M_5	$\frac{80}{246} = \frac{1}{3.07}$
M_{16}	$\frac{90}{270} = \frac{1}{3}$

$\therefore$ The least ratio is for M_5. Choice (B)

What is the sum of the highest difference and the least difference between the highest and the least scores in any of the given tournaments?

a) 293

b) 283 (Your answer - ✗)

c) 280

d) 270

Solution:

Tournament	M_1	M_2	M_3	M_4	M_5	M_6	M_7	M_8	M_9	M_{10}	M_{11}	M_{12}	M_{13}	M_{14}	M_{15}	M_{16}	M_{17}	M_{18}
Difference	155	162	154	157	166	120	100	113	124	121	162	150	160	140	150	180	170	170

From the above table, the highest difference (ie 180) occurs in M_{16} and the least difference (ie 100) occurs in M_7

$\therefore$ The required sum = 180 + 100 = 280. Choice (C)

Solution:

The sum of the lowest and the highest scores right now is 230 + 80 = 310. The maximum total score in the other two matches would be0 230 + 230 = 460 and the minimum total score would be 80 + 80 = 160.
∴ total runs scored in four matches lies between 470 and 770.
The average score would be between $\frac{470}{4}$ = 117.5 and $\frac{770}{4}$ = 192.5. Choice (B)

In how many of the given tournaments is the difference between the highest and the least scores of the team, greater than 120 but less than 140?

a) 0

b) 2 (Your answer - ✓)

c) 3

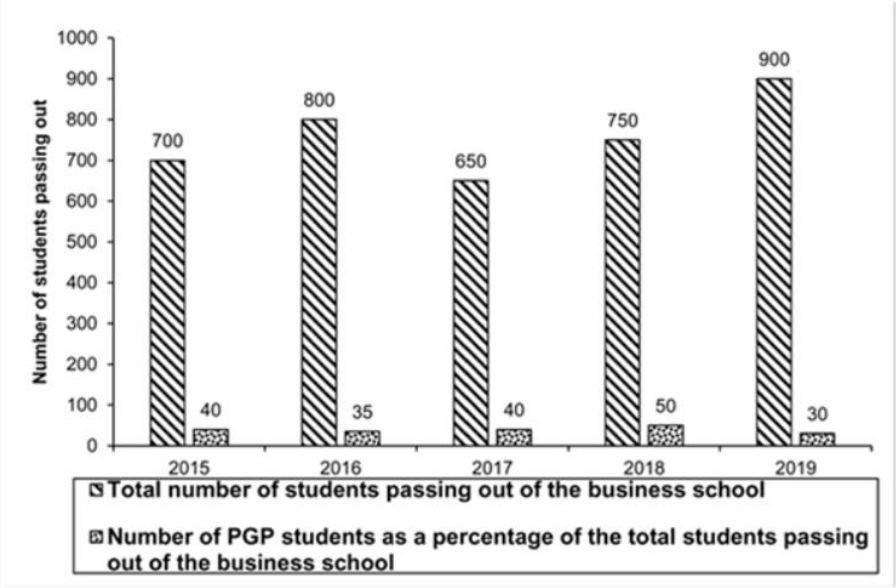
d) More than 3

Solution:

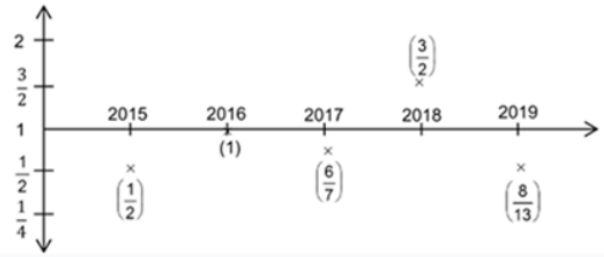
From the table, the tournaments which satisfy the criteria are M_9 and M_{10}.
Choice (B)

Directions for questions 6 to 10: Answer these questions on the basis of the information given below.

A certain business school offers the following programs: PGP, PGPM and FPM. The bar graph gives the total number of students passing out of the business school and the number of PGP students passing out as a percentage of the total students passing out of the business school in five different years.



The following line graph gives the ratio of the number of FPM to PGPM students passing out of the business school.



Directions for questions 6 to 8: Type in your answer in the input box provided below the question.

In which year was the number of FPM students passing out of the business school the second highest?

Solution:

Year	Number of Students passing out			
	Total	PGP	PGPM	FPM
2015	700	280	280	140
2016	800	280	260	260
2017	650	260	210	180
2018	750	375	150	225
2019	900	270	390	240

The number of FPM students passing out of the school is the 2nd highest in 2019.

Ans: (2019)

Directions for questions 6 to 8: Type in your answer in the input box provided below the question.

What is the difference between the total number of PGP and FPM students passing out of the business school in the given period?

Your Answer: 420 ✅

Solution:

Year	Number of Students passing out			
	Total	PGP	PGPM	FPM
2015	700	280	280	140
2016	800	280	260	260
2017	650	260	210	180
2018	750	375	150	225
2019	900	270	390	240

The total number of PGP students who passed out from the school in the given period = 280 + 280 + 260 + 375 + 270 = 1465
The total number of FPM students who passed out from the college in the given period = 140 + 260 + 180 + 225 + 240 = 1045
The required difference = 1465 – 1045 = 420 Ans: (420)

Directions for questions 6 to 8: Type in your answer in the input box provided below the question.

In which year is the number of PGPM students passing out of the business school closest to the average number of PGPM students passing out of the business school in the given period?

Solution:

Year	Number of Students passing out			
	Total	PGP	PGPM	FPM
2015	700	280	280	140
2016	800	280	260	260
2017	650	260	210	180
2018	750	375	150	225
2019	900	270	390	240

Arrange number of PGPM students who passed out of the school in the given period

$$= \frac{280+260+210+150+390}{5} = 258$$

258 is closest to 260
∴ The required year is 2016. Ans: (2016)

Directions for questions 9 and 10: Select the correct alternative from the given choices.

The number of PGP students passing out of the business school in 2015 is approximately what percentage of the number of non-PGP students passing out of the school in 2019?

a) 34.54%

b) 41.62%

c) 44.44% (Your answer - ✓)

d) 46.89%

Solution:

Year	Number of Students passing out			
	Total	PGP	PGPM	FPM
2015	700	280	280	140
2016	800	280	260	260
2017	650	260	210	180
2018	750	375	150	225
2019	900	270	390	240

The required percentage $= \frac{280}{900-270} \times 100$

$= \frac{280}{630} \times 100 \cong 44.44\%$ Choice (C)

Directions for questions 9 and 10: Select the correct alternative from the given choices.

In how many years is the number of FPM students passing out of the business school as a percentage of the number of PGP students passing out of the business school greater than 50% but less than 80%?

a) 1

b) 2 (Your answer - ✓)

c) 3

d) 4

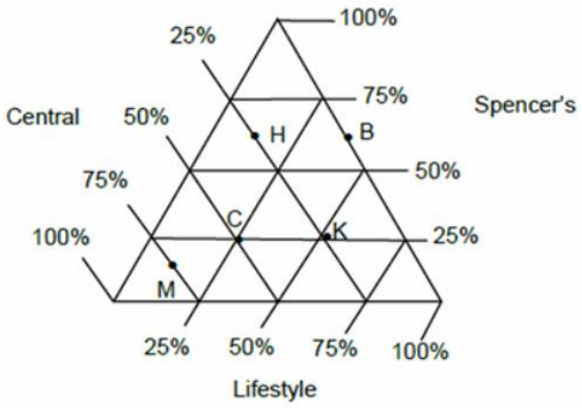
Solution:

Year	Number of Students passing out			
	Total	PGP	PGPM	FPM
2015	700	280	280	140
2016	800	280	260	260
2017	650	260	210	180
2018	750	375	150	225
2019	900	270	390	240

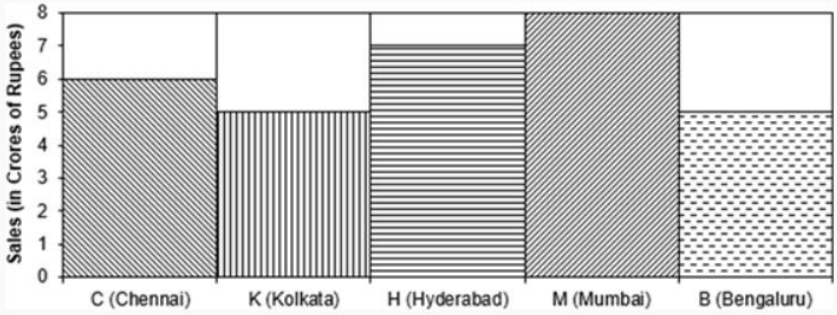
The required condition is met only in two years viz., 2017 and 2018. Choice (B)

Directions for questions 11 to 15: Answer these questions on the basis of the information given below.

The graph below shows the distribution of the percentage sales in three shopping malls – Spencer's, Life Style and Central, in five cities – C, K, H, M and B – in a particular month.



The graph below shows the total of sales (in Rs.crore) in these three shopping malls in each of the five cities for that month.



In which of the given cities is the sales of Lifestyle the maximum?

a) Hyderabad

b) Mumbai

c) Kolkata

d) Chennai

Solution:

The following table describes the information given in the triangular diagram as well as the table.

City / Malls	C		K		H	
	% sales	Sales	%sales	Sales	%sales	Sales
Spencer's	25%	1.5×10^7	25%	1.25×10^7	62.5%	4.375×10^7
Lifestyle	25%	1.5×10^7	50%	2.5×10^7	12.5%	$.875 \times 10^7$
Central	50%	3×10^7	25%	1.25×10^7	25%	1.75×10^7
Total sales (in ₹)	6×10^7		5×10^7		7×10^7	

M		B	
%sales	Sales	%sales	Sales
12.5%	1.0×10^7	62.5%	3.125×10^7
12.5%	1.0×10^7	37.5%	1.875×10^7
75%	6×10^7	0%	0
8×10^7		5×10^7	

From the table, it is clear that Lifestyle has the maximum sales in Kolkata. Choice (C)

In how many of the given cities is the sales of Spencer's more than the combined sales of the other two malls?

- a) One
- b) Two
- c) Three
- d) Four

Solution:

The following table describes the information given in the triangular diagram as well as the table.

City / Malls	C		K		H	
	% sales	Sales	%sales	Sales	%sales	Sales
Spencer's	25%	1.5×10^7	25%	1.25×10^7	62.5%	4.375×10^7
Lifestyle	25%	1.5×10^7	50%	2.5×10^7	12.5%	$.875 \times 10^7$
Central	50%	3×10^7	25%	1.25×10^7	25%	1.75×10^7
Total sales (in ₹)	6×10^7		5×10^7		7×10^7	

M		B	
%sales	Sales	%sales	Sales
12.5%	1.0×10^7	62.5%	3.125×10^7
12.5%	1.0×10^7	37.5%	1.875×10^7
75%	6×10^7	0%	0
8×10^7		5×10^7	

From the table, only in Hyderabad and Bengaluru are the sales of Spencer's more than that of the other two. Choice (B)

In which of the given cities was the sales of Spencer's and Lifestyle the same?

- a) Chennai
- b) Kolkata
- c) Hyderabad
- d) Bengaluru

Solution:

The following table describes the information given in the triangular diagram as well as the table.

City / Malls	C		K		H	
	% sales	Sales	%sales	Sales	%sales	Sales
Spencer's	25%	1.5×10^7	25%	1.25×10^7	62.5%	4.375×10^7
Lifestyle	25%	1.5×10^7	50%	2.5×10^7	12.5%	$.875 \times 10^7$
Central	50%	3×10^7	25%	1.25×10^7	25%	1.75×10^7
Total sales (in ₹)	6×10^7		5×10^7		7×10^7	

M		B	
%sales	Sales	%sales	Sales
12.5%	1.0×10^7	62.5%	3.125×10^7
12.5%	1.0×10^7	37.5%	1.875×10^7
75%	6×10^7	0%	0
8×10^7		5×10^7	

From the table, the sales of Spencer's and Lifestyle are equal only in Chennai.

Choice (A)

What is the total sales (in Rs.crores) of Central in all the given five cities?

- a) 10
- b) 11
- c) 12
- d) 14

Solution:

The following table describes the information given in the triangular diagram as well as the table.

City / Malls	C		K		H	
	% sales	Sales	%sales	Sales	%sales	Sales
Spencer's	25%	1.5×10^7	25%	1.25×10^7	62.5%	4.375×10^7
Lifestyle	25%	1.5×10^7	50%	2.5×10^7	12.5%	$.875 \times 10^7$
Central	50%	3×10^7	25%	1.25×10^7	25%	1.75×10^7
Total sales (in ₹)	6×10^7		5×10^7		7×10^7	

M		B	
%sales	Sales	%sales	Sales
12.5%	1.0×10^7	62.5%	3.125×10^7
12.5%	1.0×10^7	37.5%	1.875×10^7
75%	6×10^7	0%	0
8×10^7		5×10^7	

From the table, we observe that the total sales (in ₹crore) of Central in all the five cities C, K, H, M and B are 3, 1.25, 1.75, 6 and 0 respectively.

∴ The total amount of sales (in ₹crores) is 3 + 1.25 + 1.75 + 6 + 0 = 12. Choice (C)

In which city is the sales (in Rs.crores) of any one shopping mall more than two thirds of the sales of all the three?

- a) Kolkata
- b) Bengaluru
- c) Hyderabad
- d) Mumbai

Solution:

The following table describes the information given in the triangular diagram as well as the table.

City / Malls	C		K		H	
	% sales	Sales	%sales	Sales	%sales	Sales
Spencer's	25%	1.5×10^7	25%	1.25×10^7	62.5%	4.375×10^7
Lifestyle	25%	1.5×10^7	50%	2.5×10^7	12.5%	$.875 \times 10^7$
Central	50%	3×10^7	25%	1.25×10^7	25%	1.75×10^7
Total sales (in ₹)	6×10^7		5×10^7		7×10^7	

M		B	
%sales	Sales	%sales	Sales
12.5%	1.0×10^7	62.5%	3.125×10^7
12.5%	1.0×10^7	37.5%	1.875×10^7
75%	6×10^7	0%	0
8×10^7		5×10^7	

From the table, Mumbai has more than two thirds of its total sales from the shopping mall "Central".

Choice (D)